

Q1a.

`?crabs`

Description:

The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b.

```
data("crabs")
str(crabs)
```

```
data.frame': 200 obs. of 8 variables:
 $ sp   : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
 $ sex  : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
 $ index: int 1 2 3 4 5 6 7 8 9 10 ...
 $ FL   : num 8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
 $ RW   : num 6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
 $ CL   : num 16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
 $ CW   : num 19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
 $ BD   : num 7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

c.

```
Frequency.Table_Species<-table(crabs$sp)
Frequency.Table_Species

Frequency.Table_Sex<-table(crabs$sex)
Frequency.Table_Sex
```

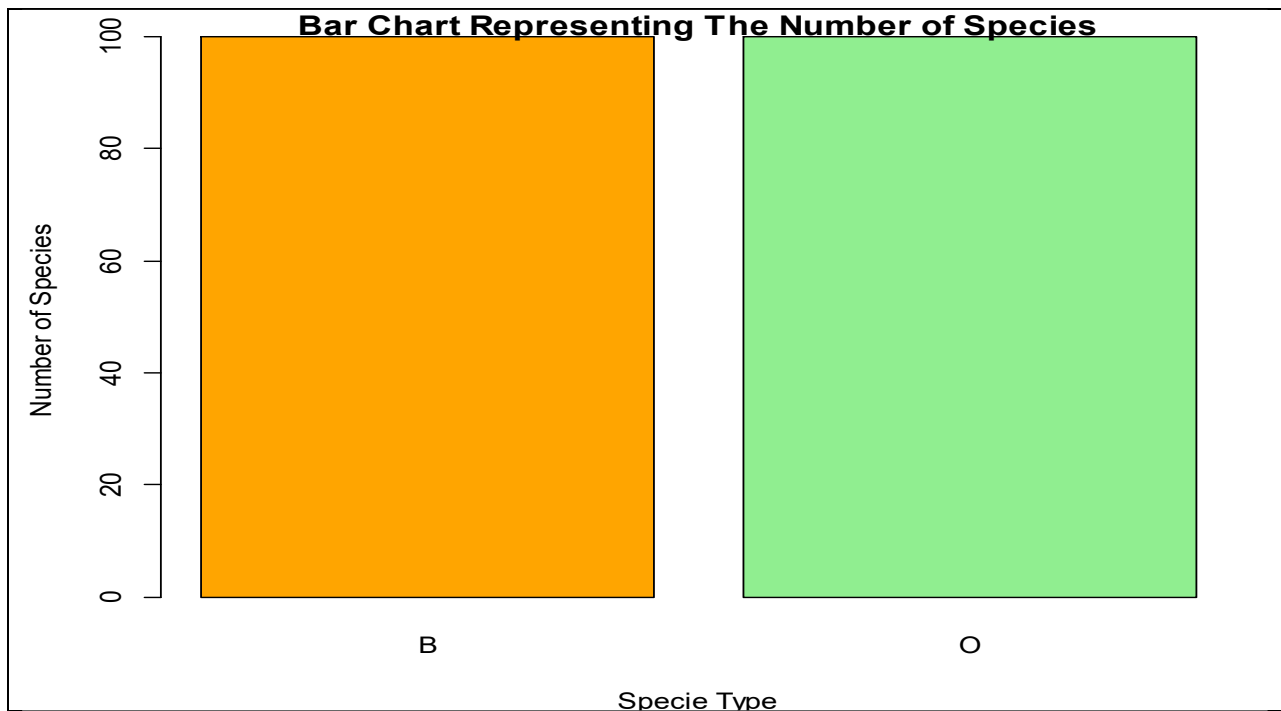
```
B   O
100 100
```

```
F   M
100 100
```

d.

```
barplot(Frequency.Table_Species,  
       main = "Bar Chart Representing The Number of Species",  
       xlab = "Specie Type",  
       ylab = "Number of Species",  
       col = c("orange", "lightgreen"))
```

FIGURE 1.1



e. Comments:

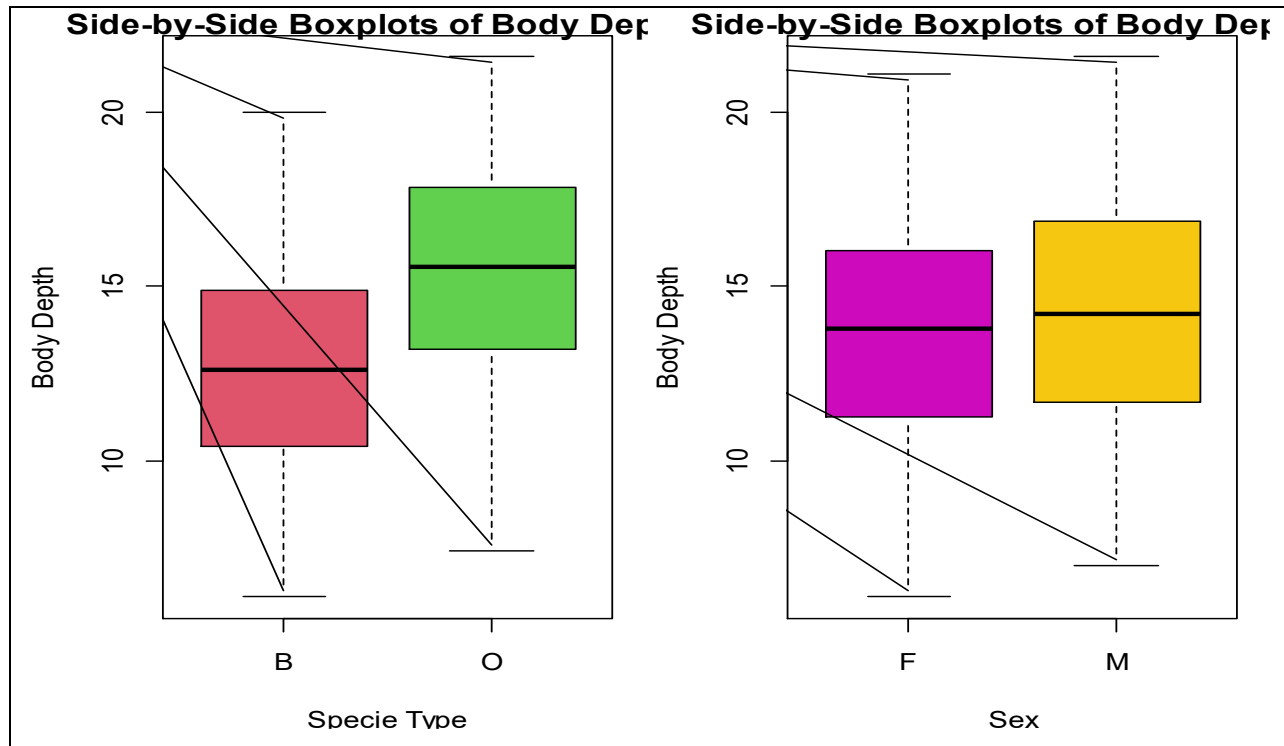
The Specie type is balanced because both groups have the same frequency.

Q2a.

```
boxplot(crabs$BD ~ crabs$sp,  
       main = "Side-by-Side Boxplots of Body Depths",  
       xlab = "Specie Type",  
       ylab = "Body Depth",  
       col = c(2:3))  
boxplot(crabs$BD ~ crabs$sex,  
       main = "Side-by-Side Boxplots of Body Depths",  
       xlab = "Sex",
```

```
ylab = "Body Depth",  
col = c(6:8))
```

FIGURE 1.2



b. Since there are no points lying outside the whiskers of the boxplots, it suggests that there are no outliers.

```

MEAN_B<-mean(crabs$BD[crabs$sp == "B"])
MEAN_O<-mean(crabs$BD[crabs$sp == "O"])
MEAN<-c(MEAN_B,MEAN_O)

Median_B<-median(crabs$BD[crabs$sp == "B"])
Median_O<-median(crabs$BD[crabs$sp == "O"])
MEDIAN<-c(Median_B,Median_O)

Mode_B<-crabs$BD[which.max(table(crabs$sp == "B"))]
Mode_O<-crabs$BD[which.max(table(crabs$sp == "O"))]
MODE<-c(Mode_B,Mode_O)
All_Species<-c("B","O")
Central_Tendencies<-data.frame(All_Species,MEAN,MEDIAN,MODE)
print(Central_Tendencies)

```

	All_Species	MEAN	MEDIAN	MODE
1	B	12.583	12.60	7
2	O	15.478	15.55	7

Q3a.

```

hist(crabs$CL[crabs$sp == "B"],
     main = "Histogram of Carapace Lenght",
     xlab = "Carapace Lenght Of Specie B",
     ylab = "Frequency",
     col = "yellow")
hist(crabs$CL[crabs$sp == "O"],
     main = "Histogram of Carapace Lenght",
     xlab = "Carapace Lenght Of Specie O",
     ylab = "Frequency",
     col = "lightgreen")

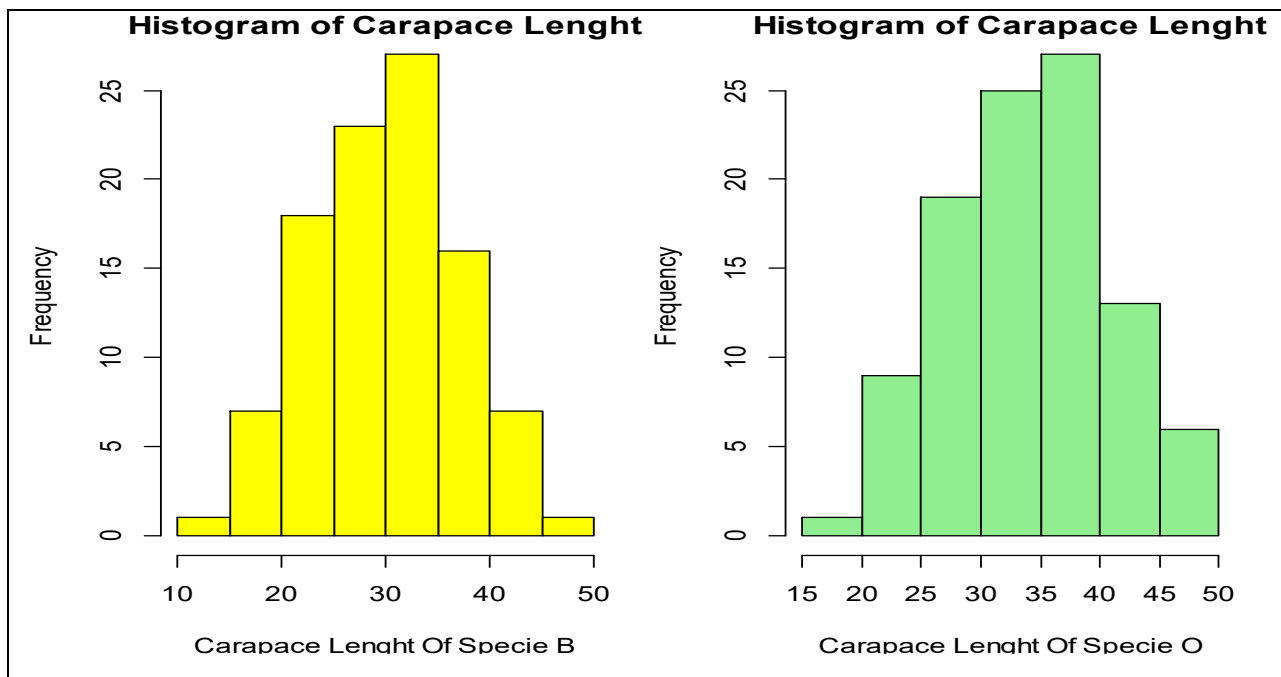
```

b.

```
describe(crabs$CL[crabs$sp == "B"])
describe(crabs$CL[crabs$sp == "O"])
```

```
describe(crabs$CL[crabs$sp == "B"])
  vars   n mean  sd median trimmed  mad min  max range skew kurtosis   se
x1     1 100 30.06 6.9  30.1   30.02 7.49 14.7 47.1  32.4 0.03   -0.65 0.69
> describe(crabs$CL[crabs$sp == "O"])
  vars   n mean  sd median trimmed  mad min  max range skew kurtosis   s
e
x1     1 100 34.15 6.76  34.55   34.29 7.56 16.7 47.6  30.9 -0.15   -0.56 0.6
8
```

Q4a.



```

MEAN_FLB<-mean(crabs$FL[crabs$sp == 'B'])
MEAN_FLO<-mean(crabs$FL[crabs$sp == 'O'])

MEAN_RWB<-mean(crabs$RW[crabs$sp == 'B'])
MEAN_RWO<-mean(crabs$RW[crabs$sp == 'O'])

MEAN_CLB<-mean(crabs$CL[crabs$sp == "B"])
MEAN_CLO<-mean(crabs$CL[crabs$sp == "O"])

MEAN_CWB<-mean(crabs$CW[crabs$sp == "B"])
MEAN_CWO<-mean(crabs$CW[crabs$sp == "O"])

MEAN_BDB<-mean(crabs$BD[crabs$sp == "B"])
MEAN_BDO<-mean(crabs$BD[crabs$sp == "O"])

MEAN_FLM<-mean(crabs$FL[crabs$sex == 'M'])
MEAN_FLF<-mean(crabs$FL[crabs$sex == 'F'])

MEAN_RWM<-mean(crabs$RW[crabs$sex == 'M'])
MEAN_RWF<-mean(crabs$RW[crabs$sex == 'F'])

MEAN_CLM<-mean(crabs$CL[crabs$sex == "M"])
MEAN_CLF<-mean(crabs$CL[crabs$sex == "F"])

MEAN_CWM<-mean(crabs$CW[crabs$sex == "M"])
MEAN_CWF<-mean(crabs$CW[crabs$sex == "F"])

MEAN_BDM<-mean(crabs$BD[crabs$sex == "M"])
MEAN_BDF<-mean(crabs$BD[crabs$sex == "F"])

MEANS_B<-c(MEAN_FLB,MEAN_RWB,MEAN_CLB,MEAN_CWB,MEAN_BDB)
MEANS_O<-c(MEAN_FLO,MEAN_RWO,MEAN_CLO,MEAN_CWO,MEAN_BDO)

MEANS_M<-c(MEAN_FLM,MEAN_RWM,MEAN_CLM,MEAN_CWM,MEAN_BDM)
MEANS_F<-c(MEAN_FLF,MEAN_RWF,MEAN_CLF,MEAN_CWF,MEAN_BDF)
Morphological_Measures<-c("FL","RW","CL","CW","BD")
tab<-
data.frame(Morphological_Measures,MEANS_B,MEANS_O,MEANS_M,MEANS_F)
print(tab)

```

Morphological_Measures		MEANS_B	MEANS_O	MEANS_M	MEANS_F
1	FL	14.056	17.110	15.734	15.432
2	RW	11.928	13.549	11.990	13.487
3	CL	30.058	34.153	32.851	31.360
4	CW	34.717	38.112	36.999	35.830
5	BD	12.583	15.478	14.337	13.724

Q5a.

```

barplot(tab)
Scatter<-pairs(crabs[4:8])

```

